

Hammad Ullah Khan

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SUMMARY

Aspiring AI/ML engineer. Passionate about AI, with a strong interest in Agentic AI, AI tools, emerging technologies, tech, entrepreneurship, and startups. Pursuing a bachelors in Artificial Intelligence from FAST. Actively involved in the AI, Computing and Data Science Society and Google Developers Club FAST.

EDUCATION

FAST NUCES, Islamabad — BS Artificial Intelligence

Jul 2024 – Aug 2028

EXPERIENCE

GREENAI AI/ML Intern

Islamabad, Pakistan Sep 2025 – Jan 2026

- Developed and trained Convolutional Neural Networks (CNNs) and regression-based machine learning models to automate crop classification, resulting in faster and more accurate agricultural analysis.
- Processed and analysed large-scale multispectral satellite and drone imagery to extract geospatial indices such as NDVI, NDWI, and BSI, enabling data-driven agricultural monitoring and improved crop insights.

NASTP AI/ML Intern

Islamabad, Pakistan Jun 2025 – Aug 2025

- Developed and tested AI models for crop type classification, crop health assessment, and yield prediction, contributing to advancements in precision agriculture across Pakistan.
- Pre-processed multispectral imagery from Sentinel-II satellites, GPSAR, and drones to generate multiband rasters for over 1,000 wheat fields, enabling accurate AI model evaluation and predictive analysis.

PROJECTS

MARS Rover 2.0

- Built and automated complex 3D environment reconstruction and object interaction by deploying YOLOv8 and Point Cloud Mapping via active SLAM on a Raspberry Pi, resulting in a fully autonomous rover capable of real-time obstacle avoidance and precise robotic arm movement and manipulation.

CROP WEED DETECTOR

- Developed an AI-powered computer vision system at GreenAI, using YOLOv5 for precise crop classification, capable of identifying 4 major weeds native to Pakistan utilizing shape, water, and vegetation indices.

UNISYNC – A WEB APP

- A centralized web application for FAST students designed to streamline university extracurricular life by integrating society and event updates. Features a digital ticketing system with automated QR code verification, club chat rooms, and secure cloud database integration to facilitate seamless student and admin engagement.

RIDEFLOW – A Smart Mobility Platform

- Developed a high-performance ride-hailing platform with comprehensive Role-Based Access Control by integrating a custom vanilla JavaScript frontend with a Supabase PostgreSQL backend, resulting in real-time ride tracking, secure digital payments, and live admin analytics.

Movie Dataset Manager

- Built and optimized a high-performance recommendation engine by parsing a 5000+ record IMDB dataset into custom STL-free C++ data structures (AVL Trees, Hash Tables, and Graphs), resulting in $O(\log n)$ search efficiency and deep BFS/DFS-based relational pathfinding.

SKILLS & INTERESTS

AI, ML & Computer Vision: Python, ML, CNNs, YOLOv8, Computer Vision, OpenCV, Scikit-learn, NumPy, Pandas, Data Scraping. **Programming & Databases:** C, C++, SQL, NoSQL, Express.js, JavaScript, HTML, CSS, Bootstrap, YAML. **Version Control:** Git, GitHub. **Core Strengths:** Leadership, Communication, Networking.

LEADERSHIP & ACTIVITIES

Directed computing events and secured major corporate sponsorships from companies like Devsinc and TeraData as Team Lead for the NASCON technology convention. During this time, I also volunteered at **The Lighthouse Orphanage**, where I mentored children and assisted with educational and extracurricular activities. Passionate and interested in AI, Automation, Geopolitics, Marathons, Polo and Squash